THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Geography Chapter 3: **Water Resources** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** hydrological cycle	All water moves in the cycle
2	**(c)** irrigated agriculture	Dry-season agriculture
3	**(a)** temples of modern India	Integrated development
4	**(c)** Sutlej–Beas river basin	Hydel power and irrigation
5	**(b)** Both A and R are true but R is not the correct explanation of A	Both true; R not the explanation
6	**(a)** Narmada	Gujarat
7	**(a)** Western Himalayas	Hill regions
8	**(d)** tankas	Rooftop rainwater harvesting
9	**(b)** Tamil Nadu	Legal provisions
10	**(a)** Meghalaya	Streams and springs

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Shortage of water in relation to needs; caused by over-exploitation, excessive use and unequal access among social groups, growing population, and pollution.	1+1
12	A barrier across flowing water that obstructs, directs or retards the flow, often creating a reservoir, lake or impoundment.	2
13	Rainwater, as it is commonly referred to in western Rajasthan; considered the purest form of natural water.	2
14	Plenty of water is now available due to the perennial Indira Gandhi Canal.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Large and growing population and greater demand; unequal access to water among social groups; over-exploitation of groundwater by industries and urban housing societies; water may be polluted by domestic and industrial wastes, chemicals, pesticides and fertilisers.	1+1+1
16	Dams where many uses of impounded water are integrated: irrigation, electricity generation, water supply for domestic and industrial uses, flood control, recreation, inland navigation and fish breeding.	1+2
17	Affect natural flow — poor sediment flow and excessive sedimentation at the bottom; rockier stream beds and poorer habitats for aquatic life; fragment rivers — difficult for fauna to migrate for spawning; reservoirs submerge vegetation and soil; induce earthquakes, water-borne diseases and pollution.	1+1+1
	OR: Hills — guls or kuls in Western Himalayas; Rajasthan — rooftop harvesting for drinking water; Bengal flood plains — inundation channels; arid areas — khadins (Jaisalmer) and johads; Bikaner, Phalodi, Barmer — underground tankas.	3
18	JJM: every rural household to get assured potable piped water at 55 litres per capita per day regularly on long-term basis; Atal Jal: in 8,220 water-stressed gram panchayats of 80 districts in seven states — brings behavioural change from consumption to conservation and smart water management.	1½+1½
	OR: Shillong: though Cherrapunjee and Mawsynram get the highest rainfall, Shillong faces shortage; nearly every household has rooftop harvesting — 15–25 per cent of household water needs. Gendathur (Mysuru, Karnataka): about 200 households installed rooftop systems; the village is rich in rainwater.	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Houses in Bikaner, Phalodi and Barmer had underground tankas for drinking water; tanks as large as a room — a house in Phalodi had a tanka 6.1 m deep, 4.27 m long, 2.44 m wide; connected to sloping roofs through a pipe; first spell of rain not collected (cleans roofs and pipes); water stored till next rainfall — reliable in summer; 'palar pani'; underground rooms adjoining the tanka kept cool; khadins and johads; now declining due to Indira Gandhi Canal.	5

	OR: Advantages: irrigation, hydel power, water supply, flood control, navigation, fish breeding, recreation; integrate agriculture with industrialisation. Disadvantages: affect natural flow and sedimentation; poor aquatic habitats; fragment rivers; submerge vegetation and soil; displacement of people; floods due to sedimentation; flood plains deprived of silt; induced earthquakes, water-borne diseases, salinisation due to change in cropping pattern.	2½+2½
20	Rooftop rainwater harvesting in rural and urban India; Gendathur (Mysuru) — 200 households, about 50,000 litres per house annually; Shillong rooftop harvesting; Tamil Nadu — compulsory with legal provisions; recharge through hand pumps and abandoned dugwells; filtration with sand and bricks; bamboo drip irrigation in Meghalaya.	5
	OR: Variations in seasonal and annual precipitation; over-exploitation; growing population; irrigation expansion for food; falling groundwater due to wells and tube-wells; industrialisation and urbanisation; pollution; need — safeguard health, ensure food security, continuation of livelihoods and productive activities, prevent degradation of ecosystems.	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Geography Chapter 3: **Water Resources** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** two billion	Global concern
2	**(b)** 11th century	Ancient hydraulic structures
3	**(d)** Allauddin Khalji	13th–14th century
4	**(d)** Ganga	First century BC
5	**(a)** Both A and R are true and R is the correct explanation of A	Bhadu song
6	**(c)** rain-fed storage structures in Rajasthan	Arid and semi-arid regions
7	**(a)** 'har khet ko pani' and 'per drop more crop'	Protective irrigation
8	**(c)** Koyna	Karnataka and AP objected
9	**(b)** 20–80 drops per minute	At the plant site
10	**(d)** Shillong	55 km away

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is continually renewed and recharged through the hydrological cycle (evaporation, condensation, precipitation).	2
12	Sringaverapura water harvesting system; dams, lakes and irrigation systems of Chandragupta Maurya's time; Bhopal Lake (11th century); Hauz Khas tank (any two).	2
13	Channels developed by people in the flood plains of Bengal to irrigate their fields.	2
14	It cleans the roofs and the pipes; only rainwater from subsequent showers is collected.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	On the basis of structure and materials — timber dams, embankment dams, masonry dams with several subtypes; on the basis of height — large and major dams, or low, medium height and high dams.	3
16	Farmers shifted to water-intensive and commercial crops; this has great ecological consequences like salinisation of soil; social disparities between rich landowners and landless poor; falling groundwater levels.	3
17	Built over the Narmada in Gujarat; covers Maharashtra, MP, Gujarat and Rajasthan; irrigation to 18.45 lakh hectares covering 3,112 villages in 15 districts of Gujarat; 2,46,000 hectares in Barmer and Jalore (Rajasthan); 37,500 hectares in Maharashtra's tribal tract through lift; drought-prone areas to become drought proof.	3
	OR: Industries are heavy users of water; they also need power, much of which comes from hydroelectric power; they discharge effluents that pollute water; urban centres with industries over-exploit groundwater.	3
18	Sedimentation in reservoirs reduces their capacity; big dams have mostly been unsuccessful in controlling floods during excessive rainfall; release of water causes floods downstream; floods cause soil erosion.	3
	OR: Bamboo pipes divert perennial springs on hilltops to lower reaches by gravity; about 18–20 litres of water enter; channel sections distribute water into branches; flow controlled by manipulating pipe positions; reduced to 20–80 drops per minute at the root of the plant.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sringaverapura (1st century BC) flood channelling of Ganga; dams, lakes and irrigation under Chandragupta Maurya; irrigation works in Kalinga (Odisha), Nagarjunakonda (AP), Bennur (Karnataka), Kolhapur (Maharashtra); Bhopal Lake (11th century); Hauz Khas tank by Allauddin Khalji for Siri Fort (13th–14th century); traditional harvesting — kuls, tankas, khadins, johads, inundation channels.	5
	OR: Launched after Independence with integrated water resources management; Nehru called them temples of modern India; integrate agriculture and village economy with industrialisation and urban growth; examples — Bhakra-Nangal (Sutlej-Beas) for hydel and irrigation; Hirakud (Mahanadi) for water conservation and flood control; Sardar Sarovar (Narmada); Damodar Valley.	5

20	Individual — avoid wastage, fix leaks, reuse water; community — rooftop rainwater harvesting, recharge of dugwells, revival of traditional systems (tankas, johads); government — Jal Jeevan Mission, Atal Bhujal Yojana, PMKSY ('per drop more crop'), mandatory rainwater harvesting (Tamil Nadu), drought-resistant crops and dry farming.	5
	OR: Ecological — poor sediment flow, excessive sedimentation, rockier stream beds, poor aquatic habitats, river fragmentation, submerged vegetation decomposition, induced earthquakes, water-borne diseases, salinisation; social — displacement of people, loss of livelihood, inter-state water disputes (Krishna-Godavari), benefits to some at the cost of others.	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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